

Production ML Workflows & Experiment Tracking

- Transition from foundational coding to industry-ready machine learning (ML) workflows.
- Focus on managing models in real-world environments.
- Build skills required for AI development and deployment roles.

Learning Objectives

- Understand the main components of production ML workflows.
- Describe the purpose of MLOps and model lifecycle management.
- Identify tools for experiment tracking; e.g., MLflow, Weights and Biases.
- Apply model documentation and versioning best practices.
- Integrate responsible AI principles when deploying models.

Why Production Workflows Matter

- Industry demands reliable, reproducible, and auditable machine learning solutions.
- Coding skills alone are not enough; need to manage models long-term.
- Skills align with standard practices in ML Engineering and DevOps roles.

Recap; Connecting Prior Learning

- Used PyTorch for model building and automation via Bash and Python.
- Set up cloud environments and storage (Weeks 7-10).
- Now, focus shifts from model creation to lifecycle and management.

What is MLOps?

- MLOps links machine learning with IT operations; similar to DevOps for software.
- Centers on automating and standardising model development, deployment, monitoring.
- Ensures quality and responsibility across ML model lifecycles.

The ML Model Lifecycle

1. Data preparation and pre-processing.
 2. Model training and validation.
 3. Experiment tracking and versioning.
 4. Model deployment and monitoring.
 5. Model retraining and retirement.
- Each stage must be documented and auditable.

Experiment Tracking; What & Why

- Experiment tracking records code, parameters, metrics, and results for each ML run.
- Enables reproducibility; you can rerun experiments and get the same results.
- Supports collaborative teamwork by sharing results and avoiding duplicated work.

Industry Tools; MLflow and Weights & Biases

- MLflow; open-source platform for managing ML experiments and models.
- Weights & Biases; cloud-based tool for in-depth experiment tracking, visualisation.
- Both support logging metrics, artifact storage, and easy comparison.

Example; Using MLflow

- Track hyperparameters, metrics, model versions.
- Simple code example;

```
import mlflow
with mlflow.start_run():
    mlflow.log_param("learning_rate", 0.01)
    mlflow.log_metric("accuracy", 0.85)
    # Save and register trained model
```

- Artifacts and logs are accessible via a web interface.

Model Versioning & Documentation

- Versioning means creating unique, traceable records for each model and experiment.
- Use naming conventions, tags, and metadata to identify model runs.
- Documentation should cover model purpose, dataset details, training procedure, and known limitations.

Best Practices; Model Lifecycle Management

- Always version datasets and source code used for training.
- Store code, parameters, metrics, and outputs for each experiment.
- Use README files or structured documentation templates for every project.

Responsible AI Principles

- Models must be monitored for fairness, bias, and consistent performance.
- Responsible AI means documenting model limitations and ethical considerations.
- Production models should trigger alerts if they behave unexpectedly.

Activity; Planning for Experiment Tracking

- Review your project code; list parameters and metrics that will be tracked.
- Choose either MLflow or Weights & Biases for experiment tracking.
- Write a short project plan for versioning and documenting models.

Industry Example; Model Tracking in Practice

- Company example; Software firm uses MLflow to track sales forecast models across regions.
- Each model run is logged with code version, dataset hash, and resulting metrics.
- This allows for quick rollback or updates when data changes.

Assessment Preparation; Why This Matters

- Accurate experiment tracking is required for portfolio assessment (Week 10).
- Model documentation supports the GPU and distributed project assessments.
- Apply these tools and principles for all future ML projects in the course.

Key Takeaways and Next Steps

- Production workflows bring accountability to ML models.
- Experiment tracking and versioning are vital for reproducibility and industry readiness.
- Prepare for next week; Autoscaling and cloud automation for ML workloads.

